Electron-coupling-to-waveguide

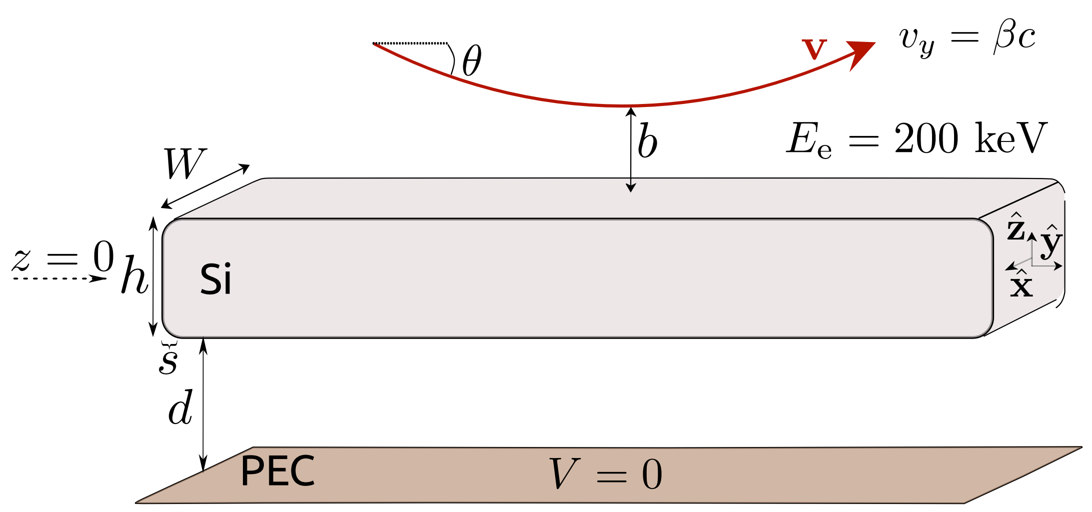
Correction of the trajectory and integration

$$\Gamma_{\mathrm{EELS}}(\omega) = 2\int_{\boxed{z_{\min}}}^{\infty} \frac{v\mathrm{d}z}{\sqrt{\left|c^2\beta^2\sin^2\theta + \dfrac{2eV(z)}{m_\mathrm{e}\gamma_\mathrm{e}}\right|}} \frac{\mathrm{d}\Gamma_{\mathrm{EELS}}(\omega, z)}{\mathrm{d}y}$$

+ electron gaussian beam and transverse energy estimation (with Cruz)

$W = 400$ nm

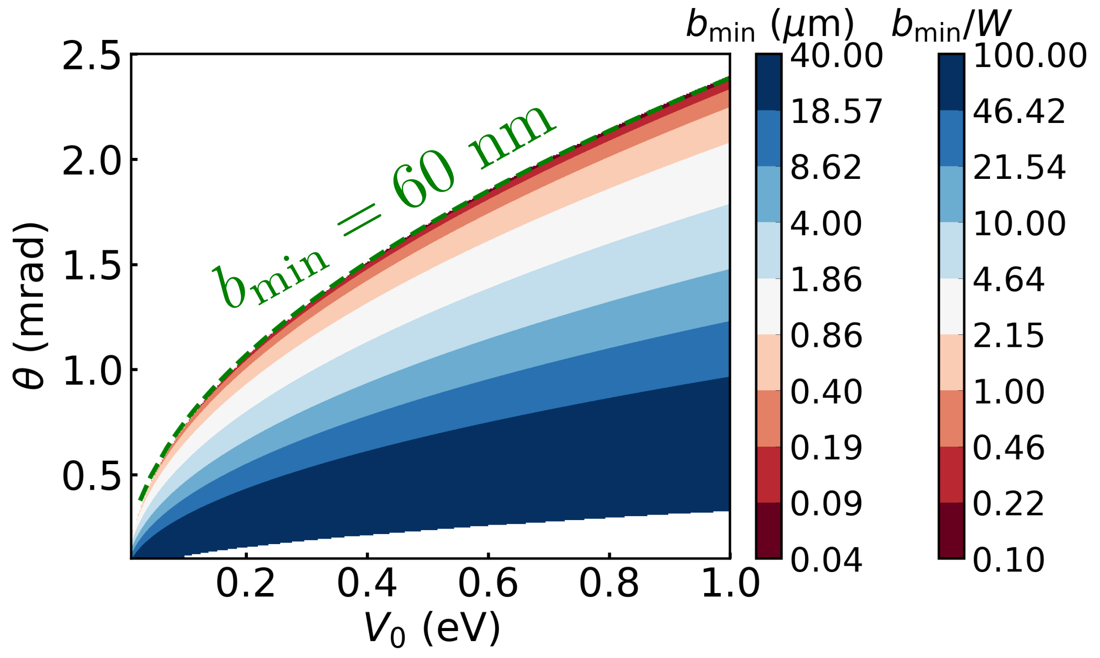
$h = 300\ \mu$m

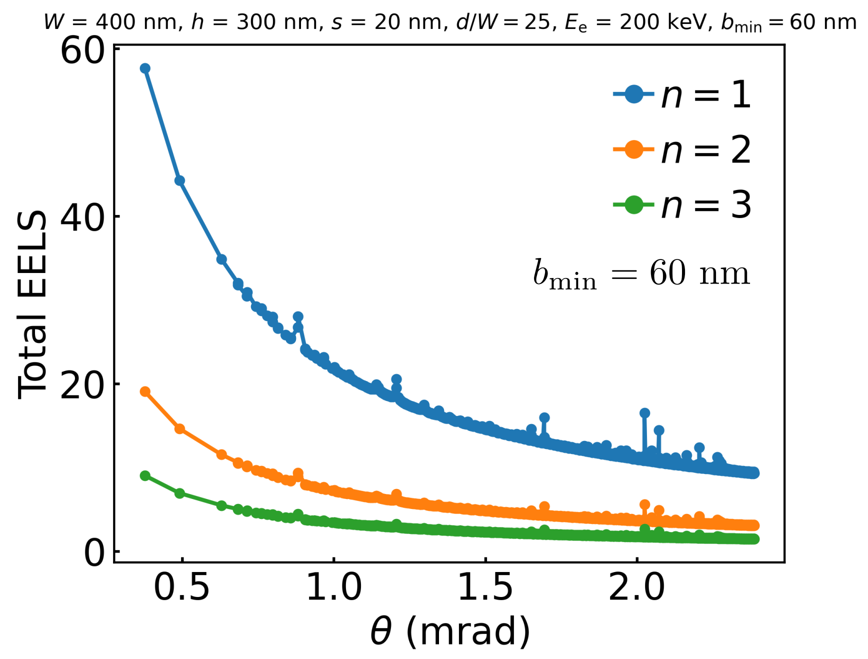


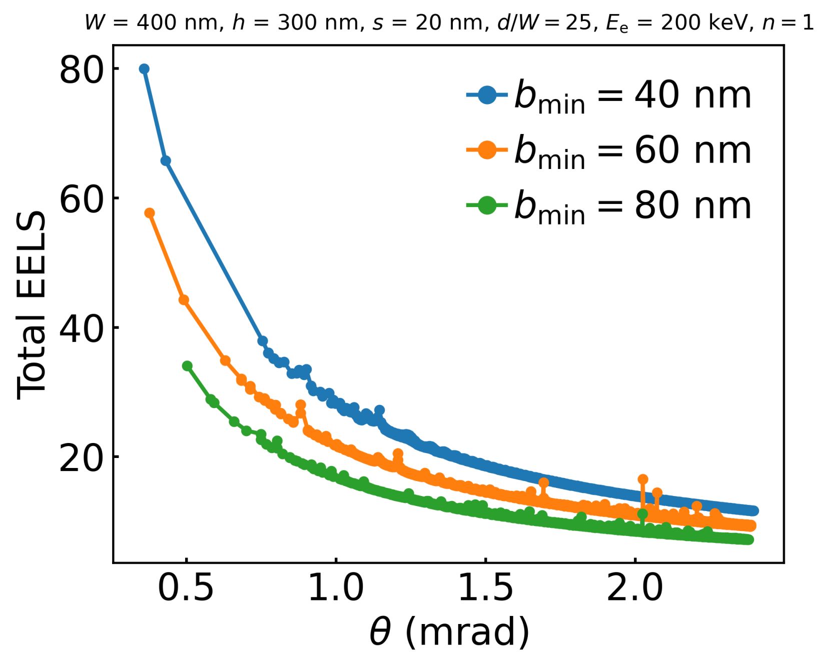
EELS for different modes for fixed b

EELS for different b for n = 1

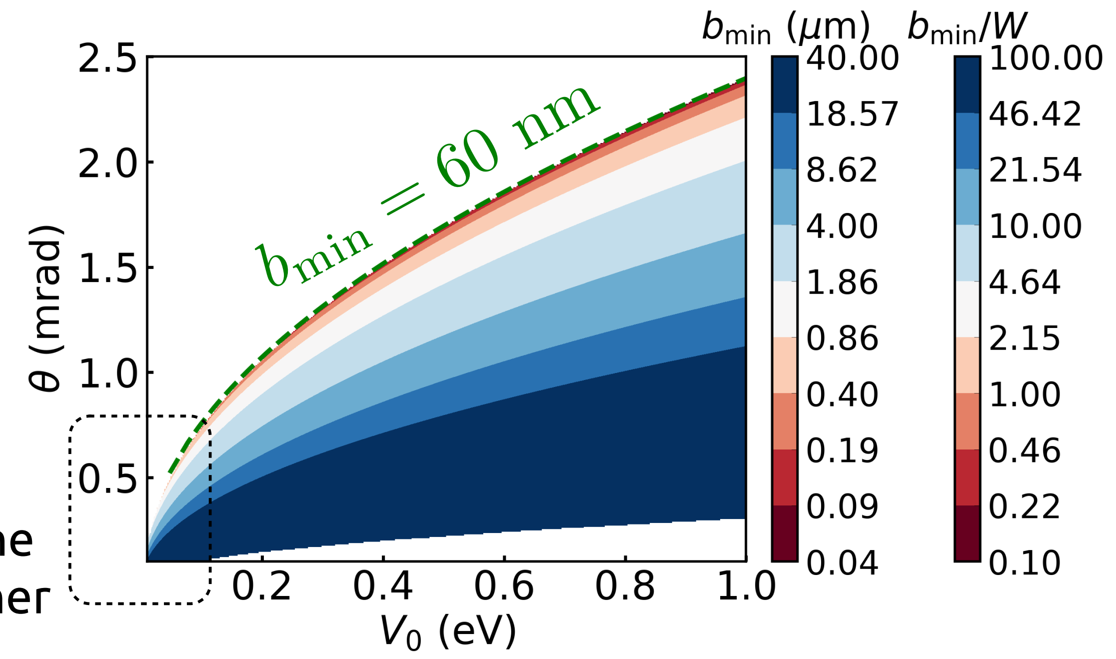
$d = 10\ \mu$m







$d = 20\ \mu$m



For bigger d, the angles are higher